47. A set of 100 students is selected randomly from an Institution. The mean height of these students is 163 *cms* and the standard deviation is 10 *cms*. Calculate the value of the test statistic under $H_0 : \mu = 167$.

48. In a random sample of 50 students from School A, 35 of them preferred junk food. In another sample of 80 from School B, 40 of them preferred junk food. Find the standard error of the difference between two sample proportions.

49. If m = 35, n = 40, $\bar{x}$ = 10.8, $\bar{y}$ =11.9, s_X = 3 and s_Y = 4, then calculate standard error of $\overline{X} - \overline{Y}$.

50. In test for population proportion, if n=500 and np=383, then calculate the value of the test statistic under $H_0 : P = 0.68$.

51. In test for two population proportions, if m = 100, n = 150, mp_X = 78 and np_Y =100, then calculate the value of the test statistic under H_0: $P_X = P_Y$.

IV. Give detailed answer to the following questions.

52. Explain the general procedure to be followed for testing of hypotheses.

53. Explain the procedure for testing hypotheses for population mean, when the population variance is unknown.

54. How will you formulate the hypotheses for testing equality of means of two populations, when the population variances are known? Describe the method.

55. Describe the procedure for testing hypotheses concerning equality of means of two populations, assuming that the population variances are unknown.

56. Give a detailed account on testing hypotheses for population proportion.

57. Explain the procedure of testing hypotheses for equality of proportion of two populations.

58. A special training programme was organized by a District Educational Officer to the VIII Standard students for improving their skill in Letter Writing. Time taken by the students in a Letter Writing competition was recorded. The average time taken by 100 randomly selected students was 15 minutes. Can the Officer decide that at 1% level of significance the mean time in this kind of exercise required to VIII Students of the district is 13 minutes, assuming the population standard deviation as 8 minutes?

59. Carry out hypotheses testing exercise for testing $H_0 : \mu_X = \mu_Y$ against $H_1 : \mu_X \neq \mu_Y$ with usual notations, when $\bar{x}$ = 7 and $\bar{y}$ =8 , σ_X = 3 and σ_Y = 2 and m = 40 and n = 40. Use α = 0.01.

60. Chief Educational Officer wanted to study the performance of XII Standard students in Mathematics subject. The following are the information obtained from randomly selected students from two Educational Districts.

Educational District	No. of students selected	Mean	Standard Deviation
A	45	62	15
B	53	60	17

Examine at 5% level of significance whether students in District A perform better compared to students in District B.

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